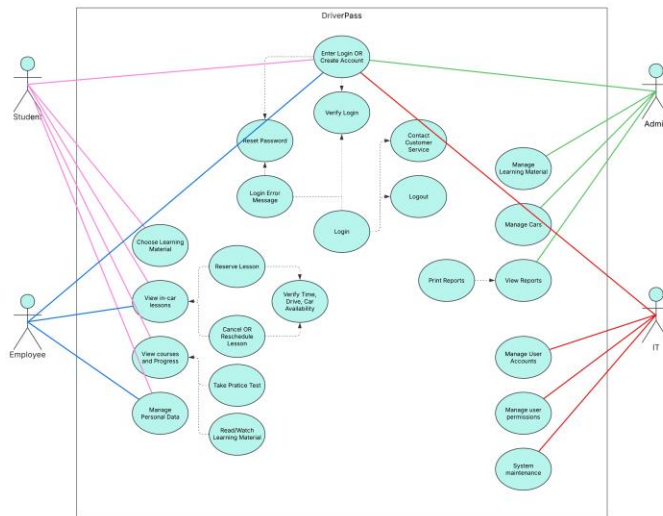


CS 255 System Design Document Template

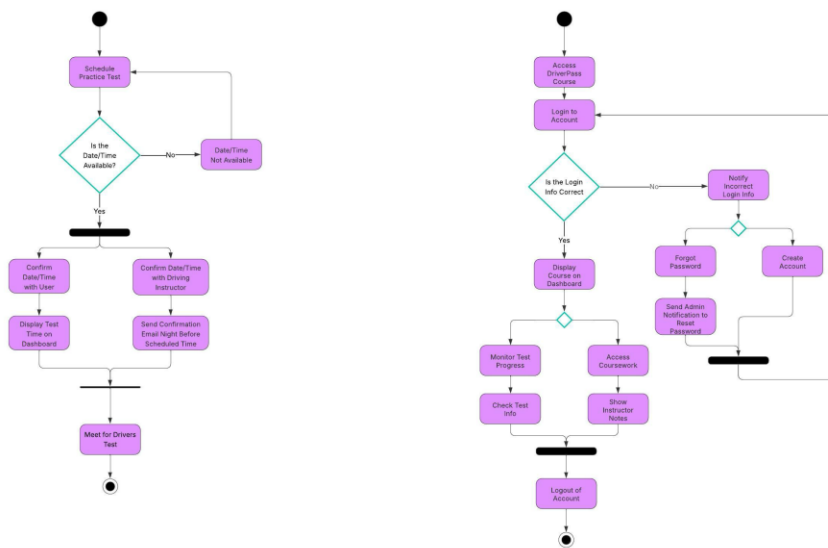
UML Diagrams

UML Use Case Diagram



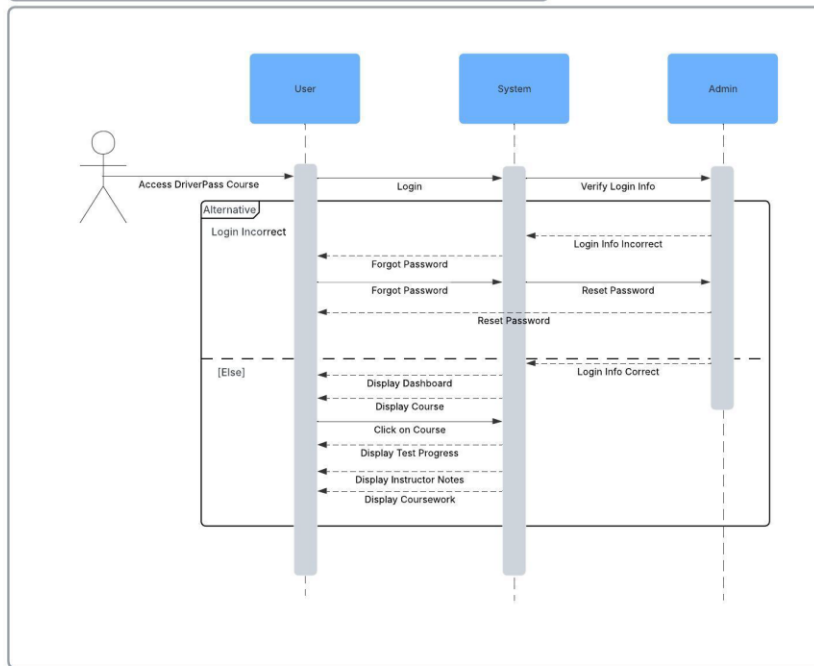
UML Activity Diagrams

DriverPass Activity Diagram



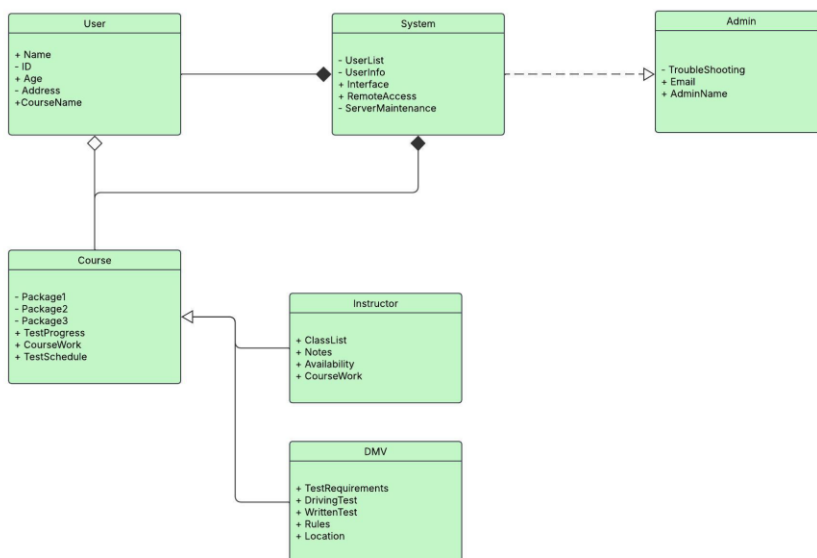
UML Sequence Diagram

DriverPass UML Sequence Diagram



UML Class Diagram

DriverPass UML Class Diagram



Technical Requirements

Hardware Requirements

- Reliable servers to host the web application and database
- Enough memory, storage, and processing power to support multiple users
- Secure internet connectivity and backup systems to protect data

Software Requirements

- Web server and database software such as MySQL
- Object-oriented programming framework for system development
- Secure authentication and HTTPS encryption to protect user information

Development Tools

- UML modeling tools like Lucidchart for system design
- IDEs such as Visual Studio Code or Eclipse
- Version control software like Git for managing updates and collaboration

Infrastructure Requirements

- Scalable and modular system design to support future growth
- Compatibility with computers, tablets, smartphones, and multiple web browsers
- Reliable uptime and system availability for users

Security and Performance Requirements

- Secure role-based access for students, instructors, and administrators
- Data protection through encryption and backup systems
- Reliable system performance and maintainability for future updates